

Education

- Oct '24 – Today **DPhil in Artificial Intelligence**, *University of Oxford*.
 - 4-year DPhil at the Oxford AIMS CDT under Prof. **Philip Torr** on **multiagent LLMs**.
 - Funded by a 250k USD Microsoft studentship. Graduation: Apr '29.
- Sep '22 – Jul '24 **MSc in Artificial Intelligence**, *University of Bologna*.
 - GPA: **29.75/30**, top of the class (170 students). Graduation grade: **110 cum laude**.
- Oct '19 – Jul '24 **Excellency Degree (COTS)**, *Institute of Superior Studies*.
 - 5-Year excellency degree (EQF8, Italian system).
 - GPA: **29.88/30**. Research topic: adversarial robustness.
- Oct '19 – Jul '22 **BSc in Computer Science**, *University of Bologna*.
 - GPA: **29.77/30**, top of the class (150 students). Graduation grade: **110 cum laude**.

Research and Teaching Experience

- Aug '25 – Now **Head**, *Institute for Decentralized AI*.
 - Research institute focused on security and infrastructure for decentralized and distributed AI systems.
 - Two main units: Oxford and Stanford.
 - Funded by the **AI Safety Fund** and the **Cosmos Institute**.
- Nov '23 – Jul '24 **Predoctoral Research Fellow**, *University of Bologna*.
 - Department fellowship on adversarial robustness under Prof. Michele Lombardi.
 - Proved that adversarial defenses for neural networks are Σ_2^P -hard (i.e., computationally infeasible).
- Mar '24 – May '24 **Visitor**, *University of Oxford*.
 - Research visit to the Department of Computer Science to conduct research on the Theory of Mind of LLMs under Dr. Emanuele La Malfa and Prof. **Michael Wooldridge**.
- Sep '22 – Jan '23 **Teaching Assistant**, *University of Bologna*.
 - Taught the Applied ML module of “Introduction to Machine Learning” for the Fall '22 semester.
- Nov '21 – Mar '22 **Machine Learning Intern**, *MindIt Srl + Kantar*.
 - Developed a **Tensorflow** LSTM implementation compatible with **IBM CPLEX** optimization algorithms.
 - Rewrote Meta's FBProphet time series model in **Tensorflow**.

Selected Publications

- Jan '25 Tobin South, **Samuele Marro**, Thomas Hardjono, Robert Mahari, Cedric Deslandes Whitney, Dazza Greenwood, Alan Chan, Alex Pentland, *Authenticated Delegation and Authorized AI Agents*, ICML 2025 (Oral).
- Oct '24 **Samuele Marro**, Emanuele La Malfa, Jesse Wright, Guohao Li, Nigel Shadbolt, Michael Wooldridge, Philip Torr, *A Scalable Communication Protocol for Networks of Large Language Models*.
- Sep '24 **Samuele Marro**, Davide Evangelista, Emanuele La Malfa, Angelo Huang, Michele Lombardi, Michael Wooldridge, *Language Models Are Implicitly Continuous*, ICLR 2025.
- Jul '23 **Samuele Marro**, Michele Lombardi. Computational Asymmetries in Robust Classification, *ICML 2023*.

Research Awards

- Aug '25 **AISF Grant**, 450k USD.
 - Grant by the AI Safety Fund (joint **Google-OpenAI-Anthropic-Microsoft** fund) on the decentralized oversight of agent networks.
- Mar '24 **Microsoft Studentship**, 250k USD.
 - Studentship for research into communication protocols between LLMs and multimodal memory in generative models.
- Jan '21 **CINECA HPC Grant**, 170k Core-Hours.
 - Grant for the Galileo100 HPC cluster for research on the computational complexity of adversarial attacks.